NUMERICAL ANALYSIS
GUIDED LAB 1

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1. Compute the absolute and relative error for

$$\sqrt{2} \approx 1 + \frac{24}{60} + \frac{51}{60^2} + \frac{10}{60^3}$$

Solution. Let's first compute the left and right hand side of the equation above using x_{approx} to represent the approximate solution (RHS). Using matlab we see that

$$\begin{aligned} x &\approx 1.41421356237, \text{ and} \\ x_{\text{approx}} &= 1.4142129630 \end{aligned}$$

Let ε_{abs} represent the absolute error, and ε_{rel} represent the relative error. Then we have the following:

$$\begin{aligned} \varepsilon_{\text{abs}} &= |\sqrt{2} - x_{\text{approx}}| \\ &= |1.41421356237 - 1.4142129630| = 5.99 \times 10^{-7} \end{aligned}$$

and we also see that:

$$\begin{aligned} \varepsilon_{\text{rel}} &= \frac{\varepsilon_{abs}}{\sqrt{2}} \\ &\approx \frac{5.99 \times 10^{-7}}{1.41421356237} \\ &\approx 4.24 \times 10_{-7} \end{aligned}$$

$\square$

2. The Newton-Raphason method for determining $\sqrt{a}$ follows from repeated iteration of

$$y_{n+1} = \frac{1}{2}\left(y_n + \frac{a}{y_n}\right) \tag{1}$$

Show that $y = \sqrt{a}$ is the fixed point of this sequence.

Proof. To achieve $y = \sqrt{a}$ as a fixed point of our equation we will need to compare the error between the n^{th} term and the $(n+1)^{\text{st}}$ term.

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We begin by subtracting $\sqrt{a}$ from both sides of the equation and performing some algebraic manipulation. Consider

$$\begin{aligned} x_{n+1} - \sqrt{a} &= \frac{1}{2}\left(x_n + \frac{a}{x_n}\right) - \sqrt{a} \\ &= \frac{x_n^2 - 2x_n\sqrt{a} + a}{2x_n} \\ &= \frac{(x_n - \sqrt{a})^2}{2x_n} \end{aligned}$$

Accordingly,

$$\begin{aligned} \lim_{n\to\infty} \frac{|x_{n+1} - \sqrt{a}|}{|x_n - \sqrt{a}} &= \lim_{n\to\infty} \frac{1}{2x_n} \\ &= \frac{1}{2\sqrt{a}} \end{aligned}$$

Hence the sequence generated by this scheme has order of convergence equal to 2 and asymptotic error constant $1/(2\sqrt{a})$. □

3. Create a script that implements Heron's algorithm (the formula in problem 1). It should take as the input: (1) the number a to square root, (2) the initial guess y_0, and (3) the total number N of iterations. In addition to outputting the approximate square root at each iteration. you may use a bulit-in functionas the "exact" answer for comparison. Plot bothe the relative error as a function of iteration for $N = 6$ on a log-log scale for $a = 2$. How sensitive is the result to y our initial guess of y_0?

Solution. First note the file problem3.m.

Consider a function that we have in file "problem3.m" that we have below:

```
function [p,absError,relError] = fixpt(a,y,n)
% Input - a, the number to sqaure root
%       - y, the initial guess
%       - n, the number of iterations to run
    p = y; % Initialize p with the initial guess
    absError = zeros(1,n);
    relError = zeros(1,n);
    for k = 1:n
        p_new = 0.5 * (p + a / p);
        absError(k) = abs(p_new - sqrt(a));
        relError(k) = absError(k) / abs(p_new);
        if absError < 1e-10
            break;
```

```
        end
        p = p_new;
    end
end

format long
[p,absError,relError] = fixpt(2,30,6)
loglog(1:6,absError, '-k')
hold on
loglog(1:6,relError,'--k')
hold off
legend("Absolute Error", "Relative Error")
```

So we see our plot from the above results as: But, if we fiddle with the initial value for the

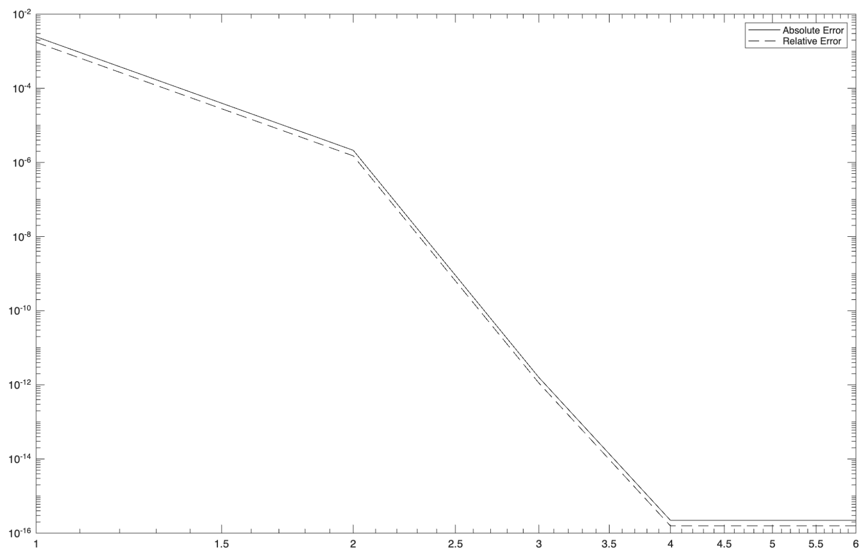

FIGURE 1. Log-Log plot of the absolute and relative error for Newton Raphson for $N = 6$, $y_0 = 1.5$

answer we see different errors. Consider Figure 2 which shows the errors when we set $y_0 = 30$ and we are done. □

4.

Solution. First note the file problem4.m.

Compute the *appoximate* relative error, and compare to the "true" errors. So we vary our algorithm a little as follows

```
function [p,absError,relError] = fixpt(a,y,n)
```

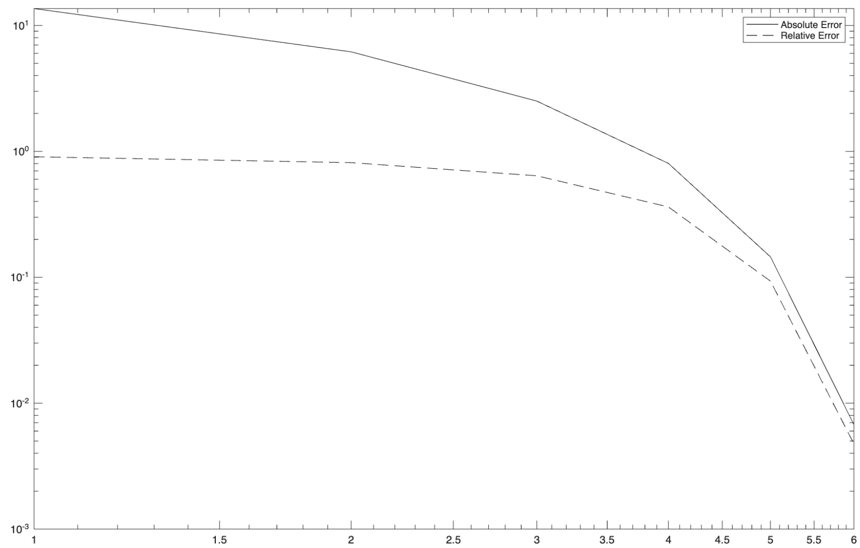

FIGURE 2. Log-Log plot of the absolute and relative error for Newton Raphson for $N = 6$, $y_0 = 30$

```
% Input - a, the number to sqaure root
%       - y, the initial guess
%       - n, the number of iterations to run
    p = y; % Initialize p with the initial guess
    absError = zeros(1,n);
    relError = zeros(1,n);
    for k = 1:n
        p_new = 0.5 * (p + a / p);
        absError(k) = abs(p_new - p);
        relError(k) = absError(k) / abs(p_new);
        if absError < 1e-10
            break;
        end
        p = p_new;
    end
end

format long
[p,absError,relError] = fixpt(2,30,6)
loglog(1:6,absError, '-k')
hold on
loglog(1:6,relError,'--k')
hold off
```

```
legend("Absolute Error", "Relative Error")
```

And for $N = 6$ and $y_0 = 30$ we see get the following errors

$$\varepsilon_6^{\text{abs}} = \begin{bmatrix} 14.966666666666667 \\ 7.450147819660015 \\ 3.659722054452168 \\ 1.706854881541622 \\ 0.657164646924791 \\ 0.138467746307099 \end{bmatrix}$$

and

$$\varepsilon_6^{\text{rel}} = \begin{bmatrix} 0.995565410199557 \\ 0.982456225846853 \\ 0.932778422047204 \\ 0.770029900059511 \\ 0.421409602464346 \\ 0.097445508110978 \end{bmatrix}$$

□

5. Plug in the equation $y_n = \sqrt{a}(1 + \delta_n)$ where $\varepsilon_n^{\text{rel}} = |\delta_n|$. Assuming that $|\delta_n| \ll 1$, simplify that to get that $\delta_{n+1} = C\delta_n^2 + \mathcal{O}(\delta_n^3)$ and specify what value the constant C should take.

Solution. Begin by substituting our δ_n formula into (1). We get the following

$$\sqrt{a}(1 + \delta_{n+1}) = \frac{1}{2}\left(\sqrt{a}(1 + \delta_n) + \frac{a}{\sqrt{a}(1 + \delta_n)}\right)$$

Now divide both sides by $\sqrt{a}$, and now notice that

$$\begin{aligned} 1 + \delta_{n+1} &= \frac{1}{2}\left(1 + \delta_n + \frac{1}{1 + \delta_n}\right) \\ &= \frac{1}{2}\left(\frac{(1 + \delta_n)^2 + 1}{1 + \delta_n}\right) \\ &= \frac{1}{2}\left(\frac{1 + 2\delta_n + \delta_n^2 + 1}{1 + \delta_n}\right) \\ &= \frac{2 + 2\delta_n + \delta_n^2}{2(1 + \delta_n)} \\ &= 1 + \frac{\delta_n^2}{2(1 + \delta_n)} \end{aligned}$$

now consider the taylor polynomial over

$$\begin{aligned}(1+\delta_n)^{-1} &= \frac{1}{\delta_n} - \left(\frac{1}{\delta_n}\right)^2 + \left(\frac{1}{\delta_n}\right)^3 - \cdots \\ &= \frac{1}{\delta_n} - \left(\frac{1}{\delta_n}\right)^2 + \mathcal{O}\left(\left(\frac{1}{\delta_n}\right)^3\right) - \cdots\end{aligned}$$

Because the sequence converges and we know that $|\delta_n| \ll 1$ we can obtain

$$\delta_{n+1} = \frac{\delta_n^2}{2} + \mathcal{O}(\delta_n^3)$$

□

6.

Solution. Problem 6 □

7.

Solution. Problem 7 □

8. Consider the problem $y_n + 1 = y_n + y_n^2 - a$. Show that $y = \sqrt{a}$ is a fixed point of our equation.

Proof. Asuming convergence fixed point c should satisify the equality $c = c + c^2 - a$ so $c = \sqrt{a}$. □

9. Implement the algorithm for the equation in problem 8, and test it with several initial guesses between 1 and 2. What do you notice about how the algorithm's convergence depends on the initial guess.

Solution. When we plot the absolute and relative as the series progresses for the different values of $y_0 = 1.2, 1.5, 1.8$ we see vastly differing errors and some that don't converge

□

10.

Solution. Problem 10 □

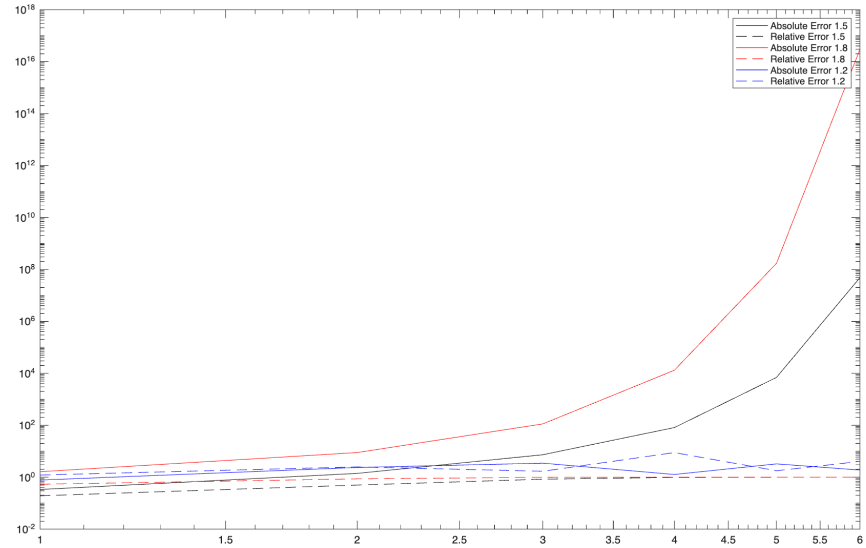

FIGURE 3. Log-Log plot of the absolute and relative error for $y_{n+1=}y_n+y_n^2-a$ with changes to initial conditions y_0